

9th Mar 2026 / DA6360

→ GD in 2 dim.

- what are the diff. & similarities
- LA results we picked up, how do they help?
- role of $\nabla^2 f$, especially in user. deriv.

- Quadratic case -

$$f(x) = \frac{1}{2}(x_1^2 + x_2^2)$$

$$\nabla f(x) = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} x_{t+1}(1) \\ x_{t+1}(2) \end{bmatrix} = \begin{bmatrix} x_t(1) \\ x_t(2) \end{bmatrix} - \alpha \begin{bmatrix} \nabla f(x_t(1)) \\ \nabla f(x_t(2)) \end{bmatrix}$$

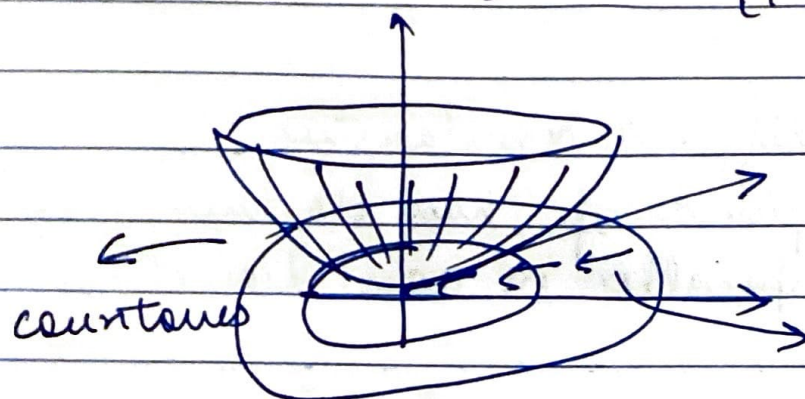
$$= \begin{bmatrix} x_t(1) \\ x_t(2) \end{bmatrix} - \alpha \begin{bmatrix} x_t(1) \\ x_t(2) \end{bmatrix}$$

$$= \begin{bmatrix} (1-\alpha) x_t(1) \\ (1-\alpha) x_t(2) \end{bmatrix}$$

$$= \begin{bmatrix} 1-\alpha & 0 \\ 0 & 1-\alpha \end{bmatrix} \begin{bmatrix} x_t(1) \\ x_t(2) \end{bmatrix}$$

$$x_t = \begin{bmatrix} (1-\alpha) & 0 \\ 0 & (1-\alpha) \end{bmatrix}^t x_0$$

$$x_t = \begin{bmatrix} (1-\alpha)^t & 0 \\ 0 & (1-\alpha)^t \end{bmatrix} x_0$$



$\alpha \in (0, 2)$
admissible
range of α

- the gradient looks
like this.

$\alpha \in (0, 1) \Rightarrow$ no oscilla^{ns}

$$- \min \frac{1}{2} (10x_1^2 + x_2^2)$$

$$\nabla f(x) = \begin{bmatrix} 10x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \end{bmatrix}$$

$$\begin{bmatrix} x_{t+1}(1) \\ x_{t+1}(2) \end{bmatrix} = \begin{bmatrix} (1-10\alpha) x_t(1) \\ (1-\alpha) x_t(2) \end{bmatrix}$$

$$= \begin{bmatrix} 1-10\alpha & 0 \\ 0 & 1-\alpha \end{bmatrix} \begin{bmatrix} x_t(1) \\ x_t(2) \end{bmatrix}$$

$$x_t = \begin{pmatrix} (1-10\alpha)^t & 0 \\ 0 & (1-\alpha)^t \end{pmatrix}$$

How do we choose α ?

$$|1-10\alpha| < 1 \quad \& \quad |1-\alpha| < 1$$

$$-1 < 1-10\alpha < 1 \quad \text{and} \quad -1 < 1-\alpha < 1$$

$$10\alpha < 2, 10\alpha > 0 \quad \& \quad \alpha < 2, \alpha > 0$$

$$\alpha \in \left(0, \frac{1}{5}\right) \quad \& \quad \alpha \in (0, 2)$$

(Tighter bound)

- To ensure convergence, $\alpha \in (0, 2/10)$

$$\alpha = \frac{2}{10}$$

$$x_t = \begin{bmatrix} (-1)^t & 0 \\ 0 & (0.8)^t \end{bmatrix} x_0$$

$$\alpha = \frac{1}{10}$$

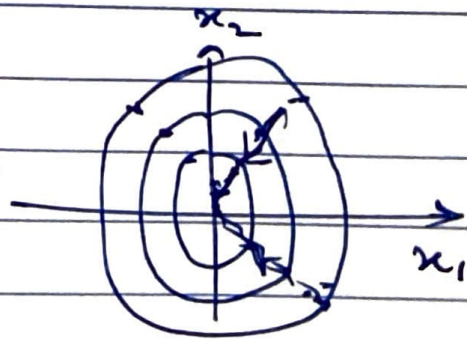
$$x_t = \begin{bmatrix} 0^t & 0 \\ 0 & (0.9)^t \end{bmatrix} x_0 \quad \begin{array}{l} \text{first coordinate} \\ \text{converges in} \\ \text{1 step} \end{array}$$

→ second one takes time.

- what if we take 2 diff. α for each coordinate (will use this later)
- we know that the speed is given by $(1-10\alpha)$ & $(1-\alpha)$

$$|1-10\alpha| = |1-\alpha|$$

$$\boxed{\alpha^* = \frac{2}{10+1} = \frac{2}{11}}$$



- what happens for 3 or more?

$$x_{t+1} = \begin{bmatrix} (1-\alpha)\lambda_1 & 0 & \dots & \dots \\ 0 & (1-\alpha)\lambda_2 & \dots & \dots \\ 0 & \dots & \dots & (1-\alpha)\lambda_d \end{bmatrix}$$

$$\lambda_1 > \lambda_2 > \lambda_3 \dots > \lambda_d > 0$$

$$\alpha_* = \frac{2}{\lambda_1 + \lambda_d} = \frac{2}{\lambda_{\max} + \lambda_{\min}}$$

